Pixels, Frames & Transformers

Handcrafted Features · Module 03

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Why Measure by Hand?

Before deep learning, computer vision meant designing measurements by hand. In social science they are often still the right tool:

Transparent
You can write the formula in your methods section.

Cheap
Seconds per thousand images on a laptop. No GPU needed.

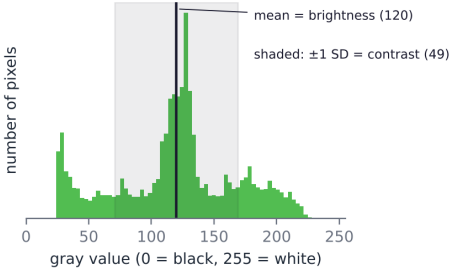
Close to a construct
Colorfulness, clutter, the size of a face: easy to link to theory.

But a precise number is not automatically a valid measure. Every feature here responds to everything in the frame, so each needs a check against what you want to measure.

The `athec` library collects dozens more such measures, with references (Peng, 2022).

Light: Brightness and Contrast





- **Brightness**: the mean gray value (0–255)
- **Contrast**: the standard deviation of the gray values
- **Clipped pixels**: the share almost black (< 30) or almost white (> 225)

On product photos, the brightest scores often come from a white background around the product, not from the package itself. The whole frame is part of the measurement.

Official portrait of Sen. Maria Cantwell, public domain, via unitedstates/images.

Color: Saturation and Colorfulness

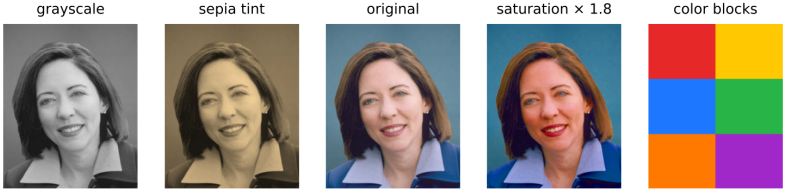
Saturation: the mean of the HSV saturation channel, how far colors are from gray.

Colorfulness (Hasler & Süsstrunk, 2003) was built to match how colorful people judge a photo to be:

$rg = R - G, \quad yb = \frac{1}{2}(R + G) - B$

$\text{colorfulness} = \sqrt{\sigma_{rg}^2 + \sigma_{yb}^2} + 0.3\sqrt{\mu_{rg}^2 + \mu_{yb}^2}$

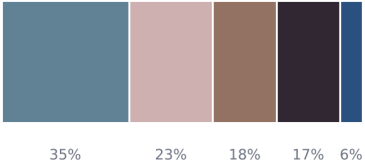
about 15 looks slightly colorful, 45 averagely, 110 extremely



The sepia tint is *more* saturated than the original but *less* colorful: the formula rewards variety, not a uniform tint.

Which Colors? Palettes with k-Means



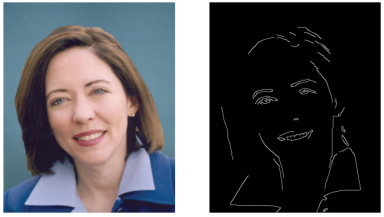


- Each pixel is a point in RGB space
- **k-means** finds k cluster centers (the dominant colors) and the share of pixels closest to each
- Cluster a downscaled copy: much faster, nearly the same result

k **is a choice.** With $k = 3$, distinct colors merge into averages that may appear nowhere in the image; with $k = 8$, a logo or a mascot's eyes get their own cluster. Like the number of topics in a topic model: report it and check that your results hold for other values.

Official portrait of Sen. Maria Cantwell, public domain, via unitedstates/images.

Complexity and Symmetry



Canny edges of the portrait

feature	what it captures	portrait
edge density	share of pixels on an edge	0.013
bytes per pixel	JPEG size at quality 75: clutter	0.084
symmetry	1 − mean difference from the mirror image	0.85

Correlated features are one finding, not two. Edge density and bytes per pixel both respond to detail; saturation and colorfulness both respond to strong colors. Check how your features correlate before you count the evidence.

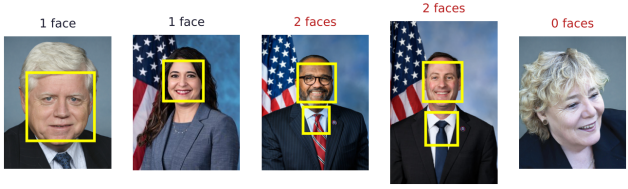
Faces with a Classic Detector

Viola–Jones (2001): slide windows of many sizes over the image; in each, test simple light-and-dark patterns that faces share.



Haar-like features: sum of light minus sum of dark. Eyes are darker than cheeks.

Fast and transparent, with well-known blind spots: faces in profile, small faces, hats, masks.



Validate where you know the answer. Every portrait shows exactly one face. On 193 portraits the detector found exactly one in **94%**; ten got extra "faces" (ties, flags), one got none.

On crowded protest photos it does much worse. Treat counts as a noisy lower bound.

Reading Text with OCR



A made-up protest sign, read by Tesseract (English and Dutch). Boxes: words kept.

image → Tesseract

↓

every word with its box and a confidence (0–100)

↓

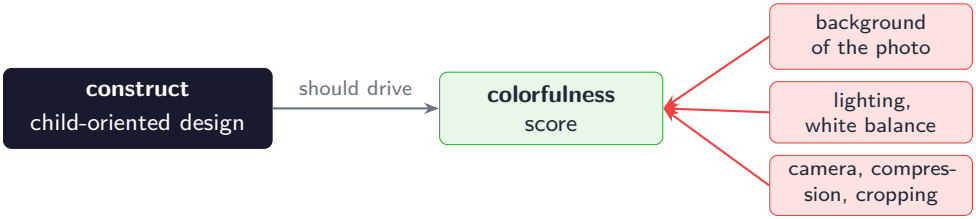
keep confidence > 60 and ≥ 2 characters

↓

features: number of words, share of the image covered by text (here 0.18), the words themselves

Results depend on the language packs you load and on resolution: Tesseract's documentation warns that accuracy drops when lowercase letters are under about 20 pixels tall.

Precise Is Not the Same as Valid



Check against known truth
like the portraits for the face detector: images where you know the right answer.

Check against human judgment
rate a sample yourself and correlate your ratings with the feature. Report it.

In the Notebook

- Light and color for breakfast cereal packaging, by Nutri-Score
- Palettes with k-means, and what k does
- Complexity, symmetry and which features measure the same thing
- Faces: validated on portraits, then tried on protest crowds
- OCR on packaging in English and Dutch
- One feature table, the input for module 04
- Your own ratings against the colorfulness score

part1_pixels/03_handcrafted_features/notebooks/03_handcrafted_features_exercises.ipynb · 7 exercises · CPU runtime